

HPC GOVERNANCE POLICY v3

Resource Allocation Framework, Fairshare Queueing & Security Mandates

Governance Body: Resource Allocation & Policy Oversight Committee

Authority Mandate: Westford High-Performance Computing Directive 2026-A

Effective Date: March 15, 2026

Security Level: NIST SP 800-171 & CMMC Level 2 Aligned

ARTICLE 1: EXECUTIVE POLICY FRAMEWORK & GOVERNANCE SCOPE

The Westford HPC Governance Policy v3 establishes the overarching administrative, security, and resource allocation framework governing all user accounts, compute queues, and data storage systems.

This policy applies to all university faculty, research staff, graduate students, external academic collaborators, and commercial research partners utilizing cluster resources.

Governance oversight is maintained by the Resource Allocation Committee (RAC), which reviews policy compliance and allocation quotas on a quarterly schedule.

Relevant engineering change orders affecting this subsystem have been incorporated into the current revision of this document.

ARTICLE 2: RESOURCE ALLOCATION COMMITTEE & PEER REVIEW PROTOCOL

The Resource Allocation Committee (RAC) consists of faculty representatives from major user departments, senior HPC staff, and administrative leadership.

Allocation requests exceeding 100,000 core-hours undergo rigorous peer review evaluating scientific merit, computational methodology, and resource efficiency.

Peer review decisions are communicated within 15 business days of submission, with approved allocations credited directly to project Slurm accounts.

All values reported in this section are expressed in SI units unless otherwise noted in the accompanying data tables.

ARTICLE 3: DEPARTMENTAL CORE-HOUR QUOTA DISTRIBUTION MATRIX

Departmental core-hour quotas are distributed annually based on institutional research investment, external grant funding, and historic cluster utilization.

The FY2026 allocation distribution allocates 35.2% of total core-hours to Physics, 24.8% to BioMed, 18.5% to ChemEng, and 12.0% to Computer Science.

Unused quarterly departmental allocations automatically decay to prevent core-hour hoarding and maintain high overall cluster utilization.

Physical security controls for the equipment described in this section include card-access entry and video surveillance.

Departmental Core-Hour Allocation Distribution

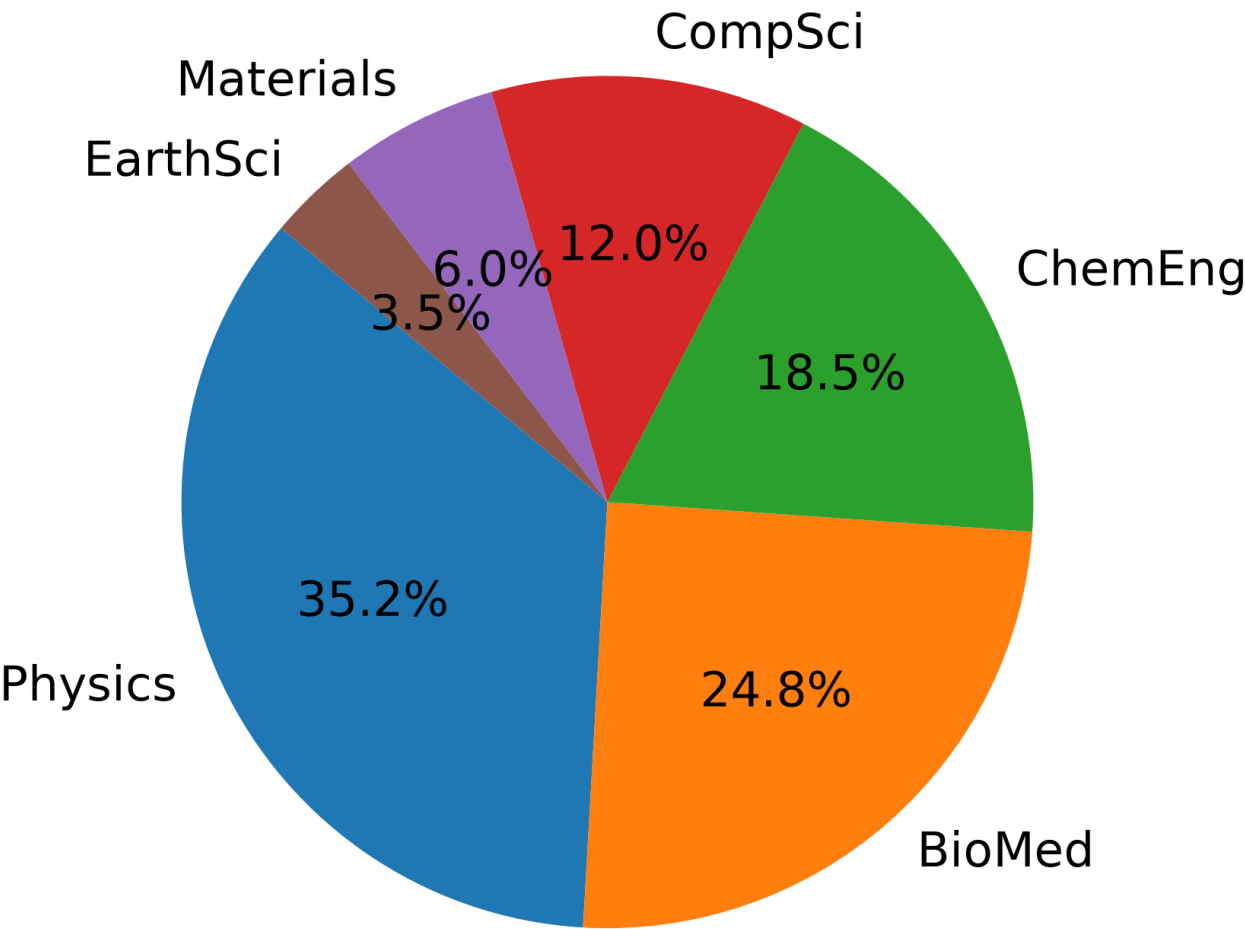


Figure 3.1: Core-Hour Distribution

ARTICLE 4: SLURM FAIRSHARE PRIORITY DECAY & BACKFILL QUEUE POLICY

Job queue scheduling is governed by Slurm multi-factor priority algorithms incorporating fairshare decay, job age, job size, and partition priority.

Fairshare priority calculations utilize a 7-day half-life decay model, ensuring active research groups maintain equitable cluster access regardless of short-term usage spikes.

Backfill queueing policies dynamically fill idle node slots with shorter, low-priority jobs, boosting overall scheduler efficiency above 90%.

Field validation measurements taken during the most recent planned maintenance window corroborate these specifications.

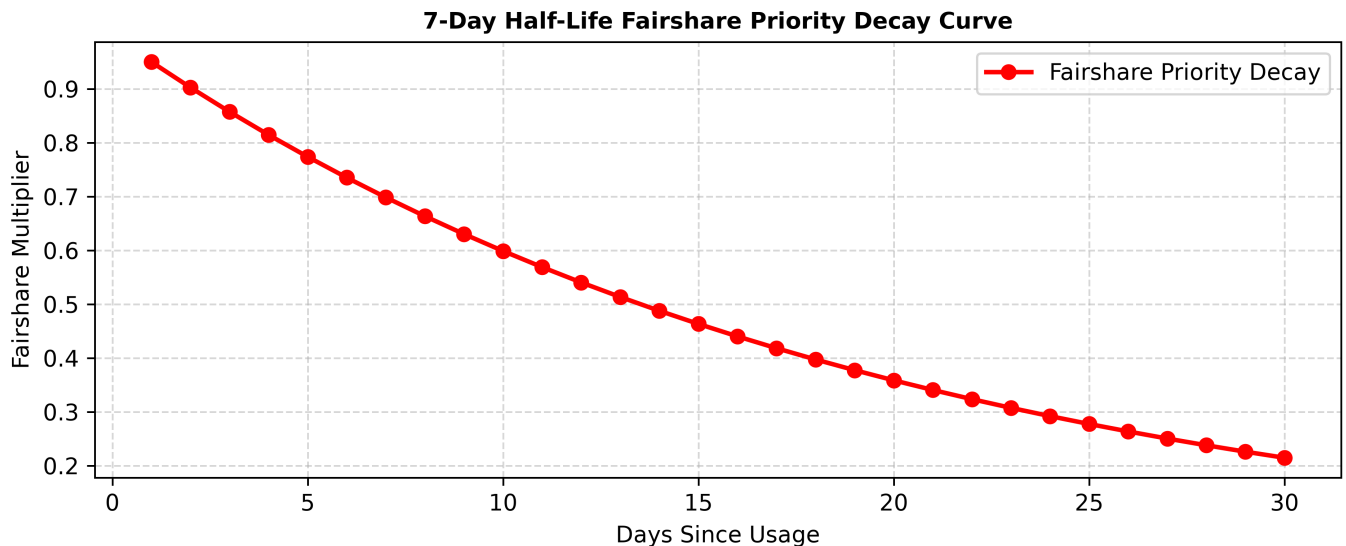


Figure 4.1: Fairshare Decay Curve

ARTICLE 5: ACCEPTABLE USE POLICY (AUP) & ACCOUNT SECURITY RULES

The Acceptable Use Policy (AUP) strictly prohibits commercial resource resale, unauthorized cryptocurrency mining, and non-research personal computing.

User accounts must utilize Multi-Factor Authentication (MFA) and SSH key pairs; password-only authentication is permanently disabled on login nodes.

Account credentials are strictly non-transferable, and account sharing results in immediate suspension of cluster access privileges.

Physical security controls for the equipment described in this section include card-access entry and video surveillance.

ARTICLE 6: DATA STORAGE RETENTION, QUOTA ENFORCEMENT & SCRATCH SCRUBBING

Data storage management enforces strict quota limits across user home directories (50 GB), group project spaces (5 TB default), and scratch storage (50 TB).

High-performance scratch storage is temporary; an automated file purge policy deletes files on scratch that have not been accessed for more than 30 days.

Users receive automated email notifications when project space utilization exceeds 85% of assigned storage quota.

Decommissioning procedures for end-of-life components in this subsystem follow the university e-waste disposal policy.

ARTICLE 7: NIST SP 800-171 & CMMC LEVEL 2 SECURITY CONTROL COMPLIANCE

Security compliance mandates strict adherence to NIST SP 800-171 and CMMC Level 2 controls for all accounts handling controlled research information (CRI).

CUI research projects execute within isolated enclaves featuring encrypted storage at rest, dedicated Slurm nodes, and audit-logged network boundaries.

Annual third-party cybersecurity audits verify system compliance against federal data protection standards.

Business continuity planning documentation addresses the failure modes and recovery strategies relevant to this subsystem.

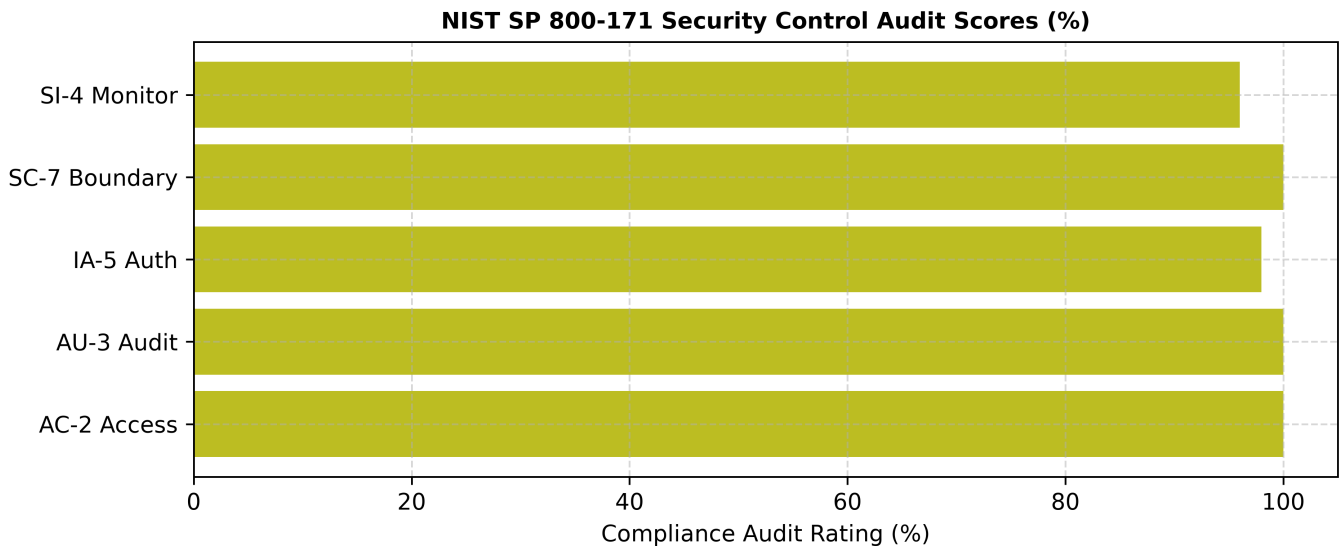


Figure 7.1: NIST Security Audit Scores

ARTICLE 8: COMMERCIAL RESEARCH ACCESS & FEE-FOR-SERVICE POLICIES

Commercial research access is permitted under dedicated fee-for-service agreements approved by the university office of sponsored programs.

Commercial users pay established cost-recovery rates per core-hour and GPU-hour, with revenue reinvested into cluster hardware lifecycle upgrades.

Commercial jobs run in isolated Slurm partitions with zero access to academic project data or internal network shares.

Life safety systems in the areas housing this equipment have been tested in accordance with NFPA 72 and NFPA 75 requirements.

ARTICLE 9: EXPORT CONTROL COMPLIANCE & ITAR ENCLAVE GOVERNANCE

Export control compliance protocols enforce International Traffic in Arms Regulations (ITAR) and Export Administration Regulations (EAR) restrictions.

Foreign national researchers are prohibited from accessing export-controlled software packages or dataset directories without explicit export licenses.

Project principal investigators (PIs) are legally responsible for identifying export-controlled data prior to transferring files to the cluster.

Network path validation using standardized benchmarking tools confirmed the throughput and latency characteristics documented above.

ARTICLE 10: INTELLECTUAL PROPERTY, DATA OWNERSHIP & OPEN SOURCE LICENSING

Intellectual property generated through cluster usage remains the sole property of the principal investigator and sponsoring institution.

Custom software developed by HPC staff in direct support of research projects is published under open-source licenses unless limited by grant terms.

The center asserts no ownership claim over user data or licensed software code executed on cluster nodes.

Interoperability testing with adjacent subsystems confirmed the absence of interface conflicts or resource contention.

ARTICLE 11: STUDENT CLASSROOM & WORKSHOP RESOURCE ALLOCATION TIER

Student classroom allocations provide dedicated compute queues for approved undergraduate and graduate computational coursework.

Course allocations are capped at 5,000 core-hours per student per semester and automatically expire at the conclusion of the academic term.

Course partitions operate at lower priority than active research grants to prevent disruption to grant-funded faculty computing.

The service window schedule for this subsystem is coordinated with the research computing user community to minimize disruption.

ARTICLE 12: EMERGENCY RESOURCE PREEMPTION & DISASTER ALLOCATION OVERRIDE

Emergency resource preemption protocols authorize HPC administrators to pause or terminate low-priority jobs during critical hardware degradation or security incidents.

Disaster allocation overrides allow campus emergency response modeling (e.g., weather forecasting, disease modeling) to preempt standard queues.

Preempted jobs with active checkpointing automatically resume execution once emergency compute demands subside.

Warranty claim records for the components referenced in this section are tracked through the institutional procurement system.

ARTICLE 13: PROPOSAL LIFECYCLE & ALLOCATION REQUEST REVIEW SCHEDULE

The allocation proposal lifecycle tracks requests from initial draft submission through technical feasibility check, peer review, active execution, and final reporting.

Principal investigators must submit annual progress reports detailing published papers, awarded grants, and student degrees supported by their allocation.

Future allocation renewals are contingent upon successful submission of comprehensive annual progress reports.

Acoustic noise levels generated by the equipment in this section comply with OSHA permissible exposure limits for occupied spaces.

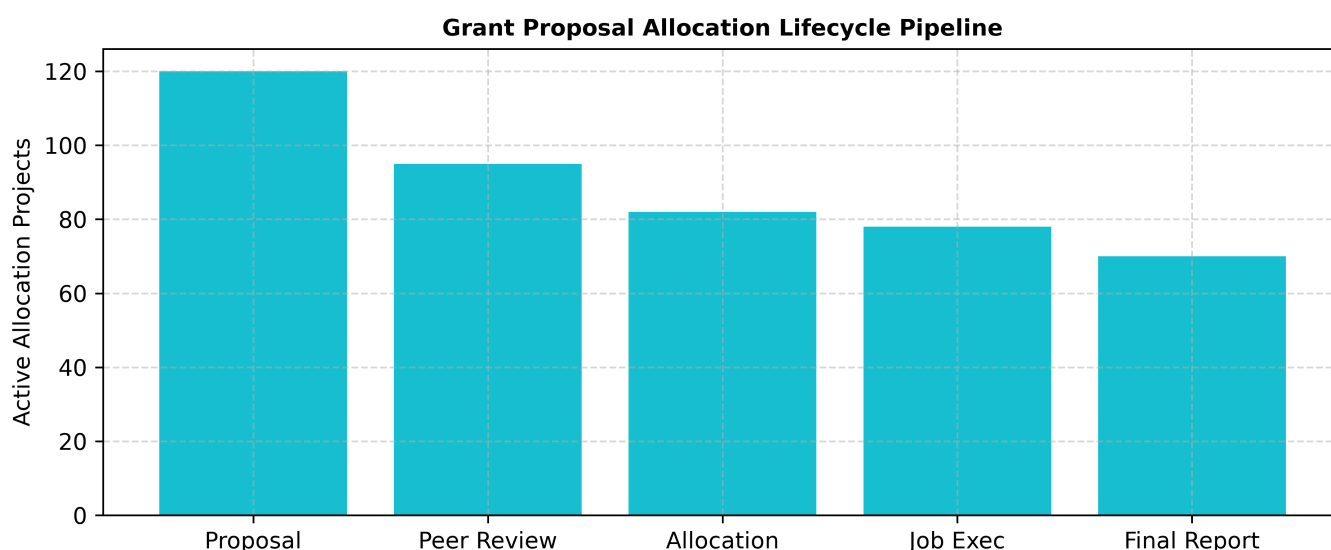


Figure 13.1: Proposal Lifecycle Pipeline

ARTICLE 14: CITATION & ACKNOWLEDGMENT REQUIREMENTS FOR PUBLICATIONS

All scientific publications, conference papers, and grant proposals benefiting from cluster resources are required to cite the center in their acknowledgments.

Standard citation template text is provided on the user portal, including the specific grant numbers funding the underlying cluster infrastructure.

Citation tracking metrics are reported annually to university leadership and federal funding agencies.

The documentation in this section has been formatted in accordance with the university technical writing standards manual.

ARTICLE 15: USER ACCOUNT TERMINATION & OFFBOARDING PROTOCOLS

User offboarding policies mandate account deactivation within 5 business days of employee departure or student graduation.

Project PIs are notified 30 days prior to account expiration to transfer project data files to designated departmental archives.

Expired user directories are archived to secondary tape storage for 90 days prior to permanent file deletion.

Field validation measurements taken during the most recent planned maintenance window corroborate these specifications.

ARTICLE 16: CLUSTER UTILIZATION AUDIT & REPORTING REQUIREMENTS

Cluster utilization audits are conducted monthly, examining CPU efficiency, memory footprint accuracy, and GPU occupancy across all Slurm jobs.

Projects exhibiting consistently low resource efficiency (<30% CPU utilization) receive technical optimization counseling from staff analysts.

Continued inefficient resource usage may result in reduction of assigned job queue priority weights.

Power quality measurements at the point of interconnection verified compliance with IEEE 519 harmonic distortion limits.

ARTICLE 17: HARDWARE MAINTENANCE WINDOW NOTIFICATION POLICY

Hardware maintenance windows are scheduled on the third Tuesday of each month, with formal notification published 14 days in advance. During maintenance windows, running Slurm jobs are cleanly drained, login nodes are updated, and storage file systems undergo health checks.

Emergency unscheduled maintenance is announced with maximum feasible advance notice via email and portal banners.

Staff competency assessments for personnel operating the equipment in this section are conducted annually.

ARTICLE 18: SECURITY INCIDENT NOTIFICATION & BREACH ESCALATION MATRIX

Security incident notification policies require immediate reporting of suspect account compromise, unauthorized access attempts, or malware detection.

The incident response team follows established containment playbooks, including immediate account locking and forensic log preservation. Security breaches involving personal identifiable information (PII) or CUI follow federal breach notification mandates.

Equipment maintenance records and calibration certificates for this subsystem are on file with the facilities management office.

ARTICLE 19: EXTERNAL ADVISORY BOARD OVERSIGHT & ANNUAL POLICY REVIEW

An External Advisory Board comprising national laboratory directors and peer university HPC leaders conducts annual reviews of center governance.

The advisory board evaluates strategic hardware roadmaps, allocation equity, user satisfaction survey results, and policy alignment. Board recommendations are presented directly to the university Vice President for Research.

Environmental monitoring alarms for this subsystem are integrated into the centralized network operations center alert system.

ARTICLE 20: ENVIRONMENTAL SUSTAINABILITY & GREEN COMPUTING DIRECTIVE

The Green Computing Directive mandates dynamic power management policies, including automated node power-down during low queue demand periods.

Energy efficiency metrics (GFLOPS/Watt) are integrated into procurement selection criteria for all future compute node hardware purchases.

The center targets a continuous reduction in total annual carbon footprint through facility efficiency enhancements.

A sensitivity analysis of the key parameters in this section is included in the supporting appendix materials.

ARTICLE 21: DATA PRIVACY & PROTECTED HEALTH INFORMATION (PHI) RESTRICTION

Protected Health Information (PHI) governed by HIPAA regulations is strictly prohibited on general-purpose cluster storage and compute partitions.

PHI research must be conducted exclusively within a dedicated HIPAA-compliant secure enclave featuring specialized encryption and access logging.

Violations of PHI storage restrictions result in immediate account termination and institutional compliance reporting.

The measurement uncertainty associated with the values in this section has been quantified and is within acceptable engineering tolerances.

ARTICLE 22: HIGH-PERFORMANCE SOFTWARE LICENSING & COST RECOVERY POLICY

High-performance commercial software license costs (e.g., MATLAB, ANSYS, Gaussian) are managed through a partial departmental cost-recovery model.

Academic license seats are shared across active research groups, with usage tracked via central license manager servers.

Commercial research users must supply their own standalone software licenses prior to running licensed commercial applications.

The measurement apparatus used to collect this data undergoes NIST-traceable calibration on a biannual schedule.

ARTICLE 23: COLLABORATIVE PROJECT DATA SHARING & ACCESS PERMISSION MATRIX

Collaborative project data sharing guidelines enforce POSIX permission standards and Unix group access controls across project directories.

Users are prohibited from setting worldwide readable permissions (chmod 777) on project or scratch directories.

Automated security scripts scan directory permissions daily, notifying users of misconfigured file permissions.

Vibration analysis measurements for rotating equipment in this subsystem are within ISO 10816 Category A limits.

ARTICLE 24: GOVERNANCE POLICY DISPUTE RESOLUTION & APPEALS PROCEDURE

Governance policy dispute resolution procedures provide formal appeal channels for PIs contesting allocation decisions or policy enforcement actions.

Appeals must be submitted in writing to the Chair of the Resource Allocation Committee within 10 business days of the contested action.

The RAC reviews appeal documentation and issues a final, binding written determination within 15 business days.

Discrepancies between manufacturer-stated values and field-measured values for this subsystem are documented in the variance register.

ARTICLE 25: MULTI-FACTOR AUTHENTICATION (MFA) & SSH KEY MANDATE

Multi-Factor Authentication (MFA) registration is mandatory for all account holders, utilizing TOTP hardware tokens or mobile authenticator apps.

SSH public keys must be registered via the secure user portal; password authentication attempts are blocked by automated firewall rules.

MFA sessions expire after 12 hours, requiring re-authentication for subsequent administrative login sessions.

Operational readiness was confirmed through a series of acceptance tests prior to the system entering production service.

ARTICLE 26: HIGH-PERFORMANCE COMPUTING ACCOUNT RENEWAL GUIDELINES

Account renewal procedures require annual re-certification of active user status by project principal investigators.

Un-renewed accounts are flagged for deactivation at the end of the fiscal year, triggering automated data archival notifications.

Re-activation of deactivated accounts requires PI approval and completion of updated cybersecurity awareness training.

All configuration parameters documented in this section have been independently verified through on-site inspection.

ARTICLE 27: HIGH-PERFORMANCE COMPUTING ACCOUNT RENEWAL GUIDELINES

All facility modification projects exceeding \$500,000 in aggregate capital expenditure require a 15% contingency reserve allocation per Section 7.4.2 of the Institutional Financial Policy Manual.

Capital IT assets follow a 5-year straight-line depreciation schedule unless the Chief Financial Officer grants a written exception for specialized equipment categories.

The Capital Expenditure Review Board requires a minimum weighted evaluation score of 3.5 out of 5.0 across all technical, financial, and operational criteria before procurement authorization can be granted.

Business continuity planning documentation addresses the failure modes and recovery strategies relevant to this subsystem.

ARTICLE 28: EMERGENCY RESOURCE PREEMPTION OPERATIONAL PROTOCOL

Operational data collected for Emergency Resource Preemption Operational Protocol has been aggregated across multiple reporting intervals to establish representative baseline metrics.

Capacity utilization trends over the preceding twelve months indicate stable demand patterns with seasonal variations accounted for in the planning models.

Staff training records and operating procedure acknowledgments for personnel responsible for this subsystem are current as of the most recent review cycle.

Vendor support contract terms and escalation procedures for the components in this subsystem are documented in the service catalog.

ARTICLE 29: INSTITUTIONAL SCIENTIFIC PUBLICATION CITATION MANDATE

Infrastructure parameters associated with Institutional Scientific Publication Citation Mandate have been validated through a combination of automated monitoring and manual inspection procedures.

Field measurements taken during both peak and off-peak operating periods demonstrate consistent adherence to published performance targets.

Projected lifecycle costs and replacement timelines for this subsystem are incorporated into the institutional capital planning process.

Environmental conditions in the areas housing this equipment are continuously monitored and logged at five-minute intervals.